

Problem 6: Parallel Optimization of the NEP Potential

June 20, 2026

1 Objective

Problem 6 asks for NEP-specific parallel optimization, including analysis of the calculation flow, optimization of atomic-density calculations, multi-element processing, memory access, correctness testing, and performance measurement over system sizes and thread counts.

This work extends the OpenMP force optimization from Problem 2 to the full NEP CPU pipeline. It targets neighbor construction, descriptor/density evaluation, mixed-element execution, and force calculation.

2 NEP Calculation Flow and Hotspots

For each molecular-dynamics step, `NEP::compute` performs:

1. periodic-boundary processing and radial/angular neighbor-list construction;
2. radial and angular atomic descriptor calculation;
3. element-dependent neural-network evaluation for atomic energy and derivatives;
4. radial, angular, and optional ZBL force/virial accumulation.

The descriptor stage is an atomic-density calculation: every center atom accumulates radial and angular contributions from its neighbors. Center atoms are independent during descriptor construction, making atom-level OpenMP parallelism natural. Force calculation is more difficult because pair contributions update two atoms and require synchronization or reduction.

The remaining NEP-specific bottleneck found in this work was the large-box cell-list neighbor search. Its expensive outer loop over center atoms was serial even though each center atom writes to a separate neighbor-list column.

3 Implementation

3.1 Large-Box Neighbor Parallelism

The outer center-atom loop in `find_neighbor_list_large_box` now uses:

```
#pragma omp parallel for schedule(static)
```

The temporary cell coordinate was moved inside the loop so every thread has a private copy. Shared cell counts and cell contents are read-only in this phase, while each thread writes only entries associated with its own center atom. Periodic-boundary coordinate conversion is also parallelized over atoms.

This optimization is especially relevant for the 512-atom benchmark, whose box is large enough to use the cell-list path.

3.2 Descriptor and Atomic-Density Parallelism

The standard and charge-aware NEP descriptor loops use explicit static scheduling. Each thread owns one center atom at a time and stores descriptor, network derivative, and angular density data at indices belonging to that atom. Model parameters and element-dependent coefficient arrays are read-only.

Static scheduling was selected because the benchmark systems have comparable local density per atom. It avoids the queue-management overhead of dynamic scheduling and preserves predictable memory access.

3.3 Multi-Element Processing

Element type is read independently for every center-neighbor pair. The OpenMP loops do not group or serialize element types, so O, Hf, and cross-element interactions execute concurrently. Correctness and performance are tested for:

- **single**: all atoms use element type 0;
- **mixed**: alternating type 0/type 1 atoms;
- **skewed**: a 3:1 ratio of type 0 to type 1.

3.4 Force and Memory Strategy

The optimized default force path uses thread-local force and virial buffers, followed by one reduction per thread. This removes atomics from the inner neighbor loop. NEP data remains in the existing structure-of-arrays layout: separate x , y , and z coordinate/force regions and neighbor-major buffers. The implementation therefore avoids interface-changing data conversions.

Thread-local reduction costs $O(TN)$ extra memory, where T is the thread count and N is the atom count. The Problem 2 serial and atomic force modes remain available for validation and memory/performance comparison.

4 Correctness Tests

The NEP OpenMP unit test now includes a mixed-element descriptor test comparing one and four threads. It also compares serial, atomic, and thread-local force paths. The dedicated Problem 6 benchmark compares every descriptor, potential, force, and virial array with a one-thread reference.

Across 27 Problem 6 benchmark cases, the maximum observed differences were:

$$\begin{aligned} \max |\Delta D| &= 0, & \max |\Delta E| &= 0, \\ \max |\Delta F| &= 7.10543 \times 10^{-15}, & \max |\Delta V| &= 4.26326 \times 10^{-14}. \end{aligned}$$

Descriptor and energy results are identical. Force and virial differences are normal floating-point reduction-order effects and are much smaller than the unit-test tolerance of 10^{-10} .

5 Benchmark Script

The dedicated files are:

```
source/source_md/tools/nep_problem6_benchmark.cpp
source/source_md/tools/run_problem6_nep_benchmark.sh
```

The command used to generate the retained CSV was:

```
cd source/source_md
tools/run_problem6_nep_benchmark.sh \
  ../../tests/PP_ORB/nep_hfo2.txt \
  64,216,512 1,2,4 single,mixed,skewed 10 \
  reports/problem6_nep_results.csv
```

The script compiles with `g++-15 -O2 -fopenmp`. Model loading is excluded from timing. Descriptor timing includes neighbor construction and atomic-density evaluation; compute timing covers the complete NEP step.

6 Performance Results

Table 1 reports four-thread speedup relative to one thread.

Atoms	Composition	Descriptor 1T (ms)	Descriptor 4T speedup	Compute 1T (ms)	Compute 4T speedup
64	Single	1.4195	3.38	3.2182	2.86
64	Mixed	1.2691	2.97	3.2470	2.94
64	Skewed	1.2723	3.06	3.2246	2.96
216	Single	3.5527	3.42	10.0926	3.43
216	Mixed	3.4916	3.31	10.2138	3.52
216	Skewed	3.5803	3.47	10.2689	3.53
512	Single	12.5353	3.68	28.4781	3.68
512	Mixed	12.7083	3.86	30.3105	4.01
512	Skewed	12.4331	3.56	29.3249	3.84

Table 1: Problem 6 NEP descriptor and full-compute scaling.

The full raw table, including 1/2/4-thread times and all numerical differences, is stored in `reports/problem6_nep_results.csv`.

7 Analysis

Descriptor/density calculation scales well for all compositions. Four-thread speedup ranges from $2.97\times$ to $3.86\times$. The nearly identical scaling for single, balanced, and skewed compositions demonstrates that element lookup and cross-element coefficient selection are thread-safe and do not introduce a serial multi-element bottleneck.

Full NEP compute reaches $2.86\times$ – $4.01\times$ speedup. The 512-atom systems exercise the newly parallelized large-box cell-list path and achieve $3.68\times$ – $4.01\times$ scaling. The mixed 512-atom case decreases from 30.3105 ms to 7.5596 ms.

The 512-atom single-element two-thread descriptor result is less efficient than the surrounding cases, likely due to runtime scheduling and measurement noise; its four-thread result returns to $3.68\times$ speedup. Repeated measurements or CPU affinity control would improve benchmark stability on a shared desktop.

8 Requirement Coverage and Conclusion

The implementation addresses the core Problem 6 requirements:

- the complete NEP flow and hotspots were analyzed;
- large-box neighbor and density preparation were parallelized;

- descriptor loops use explicit OpenMP scheduling;
- single- and multi-element systems were verified;
- memory behavior and force reductions were considered;
- correctness was checked across sizes, compositions, and threads;
- performance was measured for 27 configurations;
- a unit test, reproducible script, CSV, and report were produced.

The optimized thread-local NEP path is numerically consistent and demonstrates strong four-thread scaling. Future work could add atom blocking to reduce thread-local memory, adaptive cutoff selection, GPU density kernels, and many-core server benchmarks.